

REBECCA: Reconfigurable heterogeneous highly parallel processing platform for safe and secure AI[☆]

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ABSTRACT

An ever increasing number of applications is starting to adopt AI/ML components to unlock advanced capabilities and functionalities that are extremely hard or even not feasible at all to implement with more traditional approaches. The cost to be paid for these advancements lies in their computational requirements, which is often so significant that dictates their offloading to dedicated powerful compute systems placed in datacenter environments. This distribution of computational tasks, although convenient for a number of applications, is prohibitive for a lot of use cases (e.g. automotive and healthcare) where latency, reliability, security, privacy and other concerns pose specific restrictions and mandate that all computations are carried out either at the source of data or very close to it. This is the main driver for all efforts that try to enable AI-related computations at client devices and the edges of the network.

The EU-funded REBECCA project has emerged as part of those efforts. Its main goal is to develop a novel system-on-chip for IoT and edge systems that will be able to handle advanced AI tasks at suitable energy/power budgets. REBECCA will integrate multiple RISC-V general purpose CPU cores along with advanced AI/ML and security accelerator engines in a single package comprised of two chiplets and the capability to tightly interconnect to reconfigurable devices. The adoption of reconfigurable hardware as a key element of the REBECCA platform will provide the required flexibility to address different markets without sacrificing the energy efficiency levels it targets through the use of custom application-specific accelerators. Besides hardware components, REBECCA will also develop the required software support in terms of operating systems, hypervisors and libraries to enable the full potential of its hardware platform.

1. Introduction

AI-based systems are widely expected to revolutionize all computing application domains. Personal healthcare, smart buildings, transportation and industrial control systems, fintech solutions, marketing, etc. [1], are going through a transformation by unlocking totally new capabilities [2], not expected some years ago. To achieve, though, these new capabilities, all computing systems deployed in these applications need to be able to process a significantly heavier workload (in both

compute and memory requirements) and handle the equally increased associated energy and power budget.

Current IoT, edge and embedded devices cannot easily handle all those requirements under their operating constraints. For example, the cost, size and weight of a system may limit its practical use of high-end electronic components, additional battery capacity or cooling requirements. A popular solution to these restrictions is to offload computations in larger systems conveniently located throughout the

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computing continuum: from the edges of the network to the largest systems available at the data centers of a cloud provider.

Although this solution is a stop gap and can be convenient to overcome a series of obstacles, it raises multiple issues. Network connectivity is elevated to a critical link in the process and for several application domains (e.g. in automotive and transportation systems) its robustness cannot be guaranteed nor the communication latency can be tolerated. Furthermore, transferring sensitive/personal data over public communication infrastructure to be processed in cloud resources which may not be under direct control of the application user leads to significant reservations regarding security and privacy [3].

Issues like the aforementioned are reported as major roadblocks that prevent further penetration of AI/ML functionalities in various application areas [4]. To mitigate them, an obvious but not easy solution is to handle all computations at the device level (where data are produced or gathered) or at least at a compute infrastructure physically located near the device (edge), which still lies under user control. The means to achieve this goal is to provide electronic devices with sufficient compute capabilities to handle the AI/ML requirements of the deployed applications. The REBECCA project [5] aims to directly address these needs by proposing and designing a platform that can be deployed on the edge and provide the necessary performance without crossing power/energy boundaries that will hinder or prohibit its adoption at that level of deployment nor it will introduce barriers as a result of reduced programmability and questionable platform security.

Accomplishing this goal calls for a holistic approach, treating the problem as an overall platform that comprises of optimized hardware and software components, as well as necessary design tools to evaluate the impact of design decisions to the final outcome. As such, REBECCA is creating a series of design tools for HW/SW co-design and co-development, targeting accurate design space exploration, a custom ASIC System-on-Chip (SoC) with AI hardware accelerators to achieve the raw performance and performance per unit of energy goals and a complete software stack of operating systems, hypervisors and libraries upon which applications can be developed and deployed.

This paper focuses on the presentation of the REBECCA SoC currently under development. The SoC integrates a multi-core RISC-V general-purpose compute system, coupled with custom accelerators suitable for AI/ML tasks as well as security engines to guarantee the uncompromised execution of workloads. The SoC will be implemented as two tightly interconnected chiplets while there will be native support to connect external FPGA devices with additional application-specific compute engines for systems that have significant performance requirements and the available physical area and power budget. At the time of writing this paper, the REBECCA ASIC prototype has been completed and the project is in the late stages of providing the final design for tape-out in the Global Foundries GF22FDX process.

The structure of the paper is as follows. Section 2 sheds light on the overall REBECCA architecture, covering all major aspects of the project (hardware platform, optimized software stack and applications, and design space exploration tools). Section 3 presents the REBECCA ASIC and more specifically its main building block, the processing chiplet. All major units that comprise the chiplet are detailed and a high-level overview of their architecture and functionality is provided. Section 4 presents the current state of implementation of the platform, covering both the hardware FPGA prototypes and the initial system software. Section 5 concludes the paper.

2. The REBECCA platform for edge-AI systems

The proposed REBECCA platform is structured as a vertical stack that includes hardware components at its lowest level with system and application software built on top of them. Spanning through the whole stack, design and modeling tools are used to facilitate the development of the aforementioned components. The overall architecture is depicted in Fig. 1.

The cornerstone of the overall platform is the novel REBECCA ASIC. The chip consists of two tightly coupled identical chiplets connected through an interposer and custom chip-to-chip interfaces. Each chiplet (termed as Processing Chiplets in Fig. 1) is built around a quad-core RISC-V processor connected with a number of custom accelerators. An internal network-on-chip (based on AXI4) manages all communications and data transfers between the general-purpose domain (RISC-V processor cores), the accelerators, and the main peripherals.

The accelerator engines primarily target the execution of AI-related tasks and include a Neuromorphic AI accelerator, a programmable array AI accelerator, a DNN accelerator and a Near-Memory Compute (NMC) component. For applications that require additional compute and memory capabilities, there is a provision to closely attach reconfigurable devices (FPGAs) through a custom chip-to-FPGA interface. The inclusion of such a large array of different AI accelerators enables application designers to use the most appropriate and efficient ones for their specific requirements. For example, Neuromorphic accelerators are orders of magnitude more efficient (in terms of inferences/second and inferences/watt) than conventional ML accelerators for numerous AI algorithms, while reconfigurable hardware allows the acceleration of applications from different domains, not only limited to vision, audio, etc.

As such, the approach of REBECCA differs from typical modern edge-AI solutions that rely on a single form of accelerator engine, usually under the general term Neural Processing Unit (NPU). REBECCA focuses on using the most appropriate form of hardware acceleration of each software domain instead of trying to build a single engine that can provide the highest operations per second in order to cover the widest array of applications. This way a more balanced and effective path towards achieving a desired performance per energy/power unit can be achieved.

Beyond AI accelerators, the system includes custom IP blocks to monitor system activity and flag potentially malicious or unwanted events at the hardware level. These secure engines provide a special purpose Intrusion Detection System (IDS) that oversees specific memory areas and communication exchanges. Both the security accelerator and the NMC module can directly access the main memory resources attached to the chip. Considering the performance requirements of the REBECCA chip in correlation with the physical area and power restrictions of the targeted embedded/IoT applications, the chip supports the use of fast DRAM through HyperRAM modules. These provide adequate memory resources in terms of overall capacity and performance while keeping the design simple and extremely compact. Lastly, to provide access to larger pools of non-volatile storage, the chip integrates a common SD card controller. A more detailed view of the organization and high-level architecture of the REBECCA processing chiplet is depicted in Fig. 2.

The extended use of specialized hardware accelerators in a system can potentially lead in software support issues as well as increased development effort for application providers. This is the reason why REBECCA is provided as a platform that includes besides the actual ASIC, critical software components: system software (firmware, operating systems and drivers), hypervisors, libraries and runtime systems that also include support of common AI frameworks.

More specifically, the project currently provides support for two operating systems that have been ported to the platform. The first one is a typical embedded linux system aimed at general purpose development for non security and safety critical applications. The second one, PikeOS [6] from SYSGO, is microkernel-based RTOS certified for Common Criteria, ISO 26262, IEC 61508, EN 50128, and DO-178. PikeOS offers a separation kernel-based hypervisor with multiple partitions for many other operating systems and applications and on top of it, another embedded linux (ElinOS [7]) is made available.

In terms of supported hypervisors, the REBECCA platform also supports the XtratuM Next Generation (XNG or simply XtratuM) hypervisor [8]. XtratuM is an embedded ECSS category B qualified type

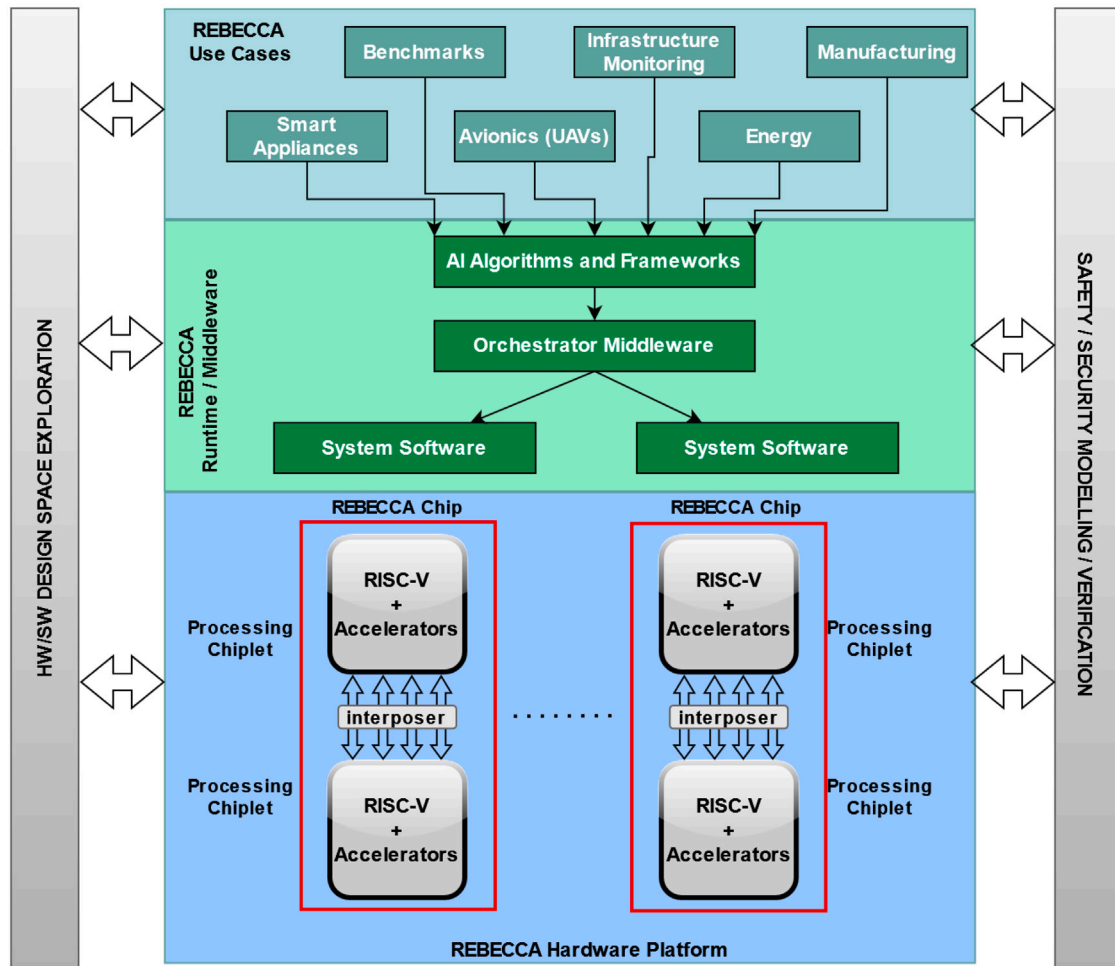


Fig. 1. The REBECCA platform.

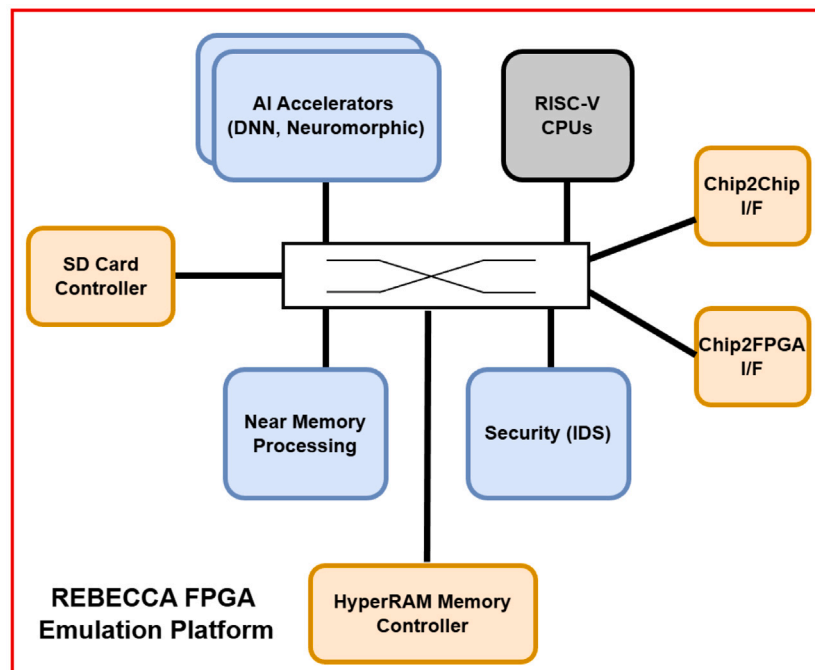


Fig. 2. The REBECCA processing chiplet high-level hardware architecture.

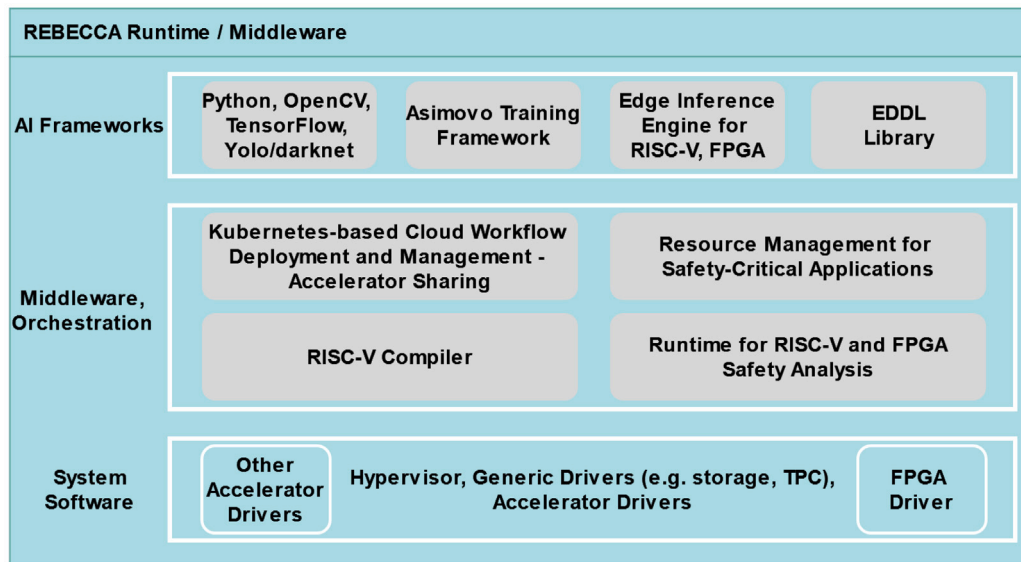


Fig. 3. The REBECCA software stack.

1 hypervisor aimed at safe and efficient embedded real-time systems through virtualization, allowing different applications to share the same hardware platform through the isolation in time and space of these applications in partitions (TSP). Thanks to XtratuM, developers can build bare-metal applications or use different operating systems over the same platform, as well as to allocate applications with different criticality levels to each partition. The use of XtratuM is convenient for systems in which the main challenges deal with safety, the reuse of certified applications, dynamic software updates and size, weight, power and cost reduction of embedded safety-critical systems.

Besides the OSES and the Hypervisor, REBECCA is also developing a runtime to enable the scheduling and management of tasks targeting the different accelerators, as well as porting and optimizing AI libraries, such as EDDL [9] and SENECA Network Builder (software library to deploy neural networks on the SENECA hardware accelerator), and AI frameworks, such as KlepsydraAI [10]. The comprehensive software stack of REBECCA is shown in Fig. 3.

Finally, the REBECCA platform includes a novel HW/SW Design Space Exploration (DSE) tool aiming to assist the development of highly efficient systems using the REBECCA components. The DSE tool will enable a designer to experiment with the cost/performance/power consumption/security/safety of different design points from the very beginning of the design phase. This will not only allow for more efficient edge-AI systems, but also for significantly faster development times. REBECCA will additionally provide the means for safety and security modeling and verification for the developed HW and SW from the very early stages of the design phase.

2.1. Applications

At this point, it is useful to provide some information on the applications that the REBECCA platform targets. This can be clearly elaborated by detailing the chosen use cases that the REBECCA project has selected to validate and demonstrate its platform.

2.1.1. Real-time defect detection in PV panels on unmanned aerial vehicles (UAV) devices

Solar power plants require reliable condition monitoring systems to ensure their productivity over time. The REBECCA platform will be integrated in a system that will perform automatic detection of anomalies using aerial Infrared (IR) and RGB images of PV panels collected on-site by UAVs. The system will be deployed on a configurable board with limited memory and power resources and will be installed on UAVs.

2.1.2. AI-powered fridges that recognize food with image recognition

The REBECCA platform will be used in the design of new smart appliances, including new electronic board designs, edge AI software designs, and overall edge AI application activities specific to connected white goods. In the specific paradigm, an AI application that recognizes food within a smart fridge will be implemented in order to provide a number of functionalities (e.g. automatic ordering of goods).

2.1.3. Water infrastructure inspection

The REBECCA platform will be integrated in robots that can vertically navigate waterbased pillars ("Poleclimber") and robots that can autonomously perform a wall inspection by hovering close to the wall ("wall inspector"). By using the REBECCA-based AIoT application the goal is to gather video as well as other sensor information in a cost effective way and with precise positioning and these data will be filtered so as to be used in high quality imagery algorithms while also being available in a 3D format useful for automatic classification of problems.

2.1.4. High-speed damage inspection and vision in the loop on semiconductor equipment inspection

Current chiplet placement and bonding techniques require very fast (significantly less than 10 ms) visual inspections. Moreover, more intelligent and numerous product inspections are increasingly demanded by applications and customers. REBECCA platform will be used to design a scalable and distributed vision architecture.

3. The REBECCA ASIC

The foundational building block of the overall REBECCA platform is the Processing chiplet. In this section, the main IP blocks that make up the chiplet are presented along with their high-level architectural details. It should be stressed again that the REBECCA ASIC is built by integrating two identical chiplets through an interposer.

3.1. General purpose processor subsystem

The REBECCA hardware system is built around a quad-core RISC-V general-purpose processor. The RISC-V ISA was chosen by the project in part due to its open nature and partly because of its modular approach. The former serves the open nature of the project while the latter enables the optimization of the overall design according to its specific needs as

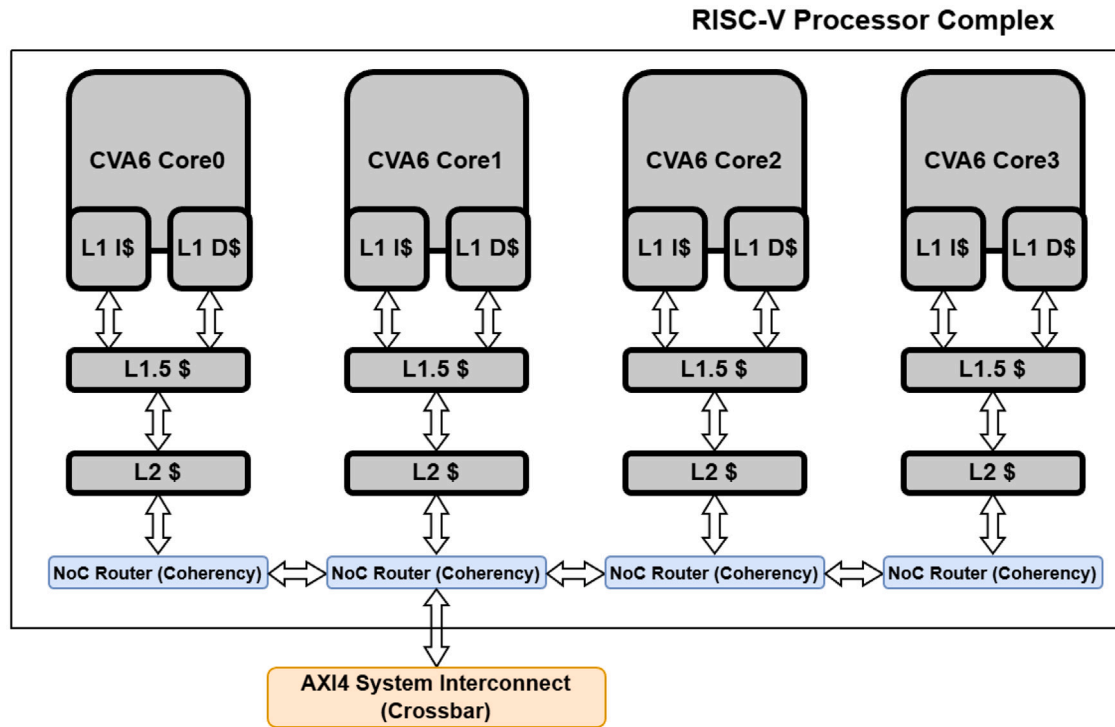


Fig. 4. The RISC-V processor complex within the REBECCA chip.

it allows for the implementation only of relevant subsets of the ISA thus enabling the proper sizing of the processor IP.

For the first iteration of the REBECCA chip, the project has opted to use a well-tested and silicon-proven open source core. More specifically, the chosen core is OpenHWGroup's CVA6 [11], which is a 6-stage, single-issue, in-order CPU that implements the 64-bit RV64GC instruction set. It provides support for all the necessary features required to boot a full-fledged OS like Linux.

Each chiplet in the REBECCA chip, integrates four of those cores in a coherent image. The coherency fabric is based on the OpenPiton framework [12], which supports the CVA6 core (among others). The framework establishes a memory subsystem that maintains cache coherence using a P-Mesh coherence protocol. The coherency is established between the Level1.5 and Level 2 caches using three P-Mesh NoCs (Fig. 4). OpenPiton can scale to support vast numbers of processor cores in a single coherent system and this ensures that future expansions of the REBECCA platform can be implemented with minimal changes.

OpenPiton's interfaces with the components outside the processor complex have been modified, in order to support the addition of multiple memory-mapped accelerators. As such, the OpenPiton NoC is interfaced with the main chip NoC that enables connectivity and data transfers between the processor complex, the main memory resources and the accelerators.

3.2. System interconnect

The main chip NoC is implemented as an AXI4 compliant [13] crossbar. The processor complex along with all accelerators and major peripherals in the chip are endpoints in this interconnect and they all access the memory resources through this NoC, as well as external devices such as additional chiplet and external FPGA devices.

As a standalone IP, the central interconnect allows a configurable number of AXI4 Masters to communicate with a configurable number of AXI4 Slaves. More specifically, the interconnect allows for:

- a configurable number of Slave and Master interfaces

- a configurable address space for each Slave
- configurable connectivity between masters and slaves to reduce unnecessary resources in case all-to-all communication is not required
- multiple simultaneous transactions in the absence of conflicts
- in case of conflicts, a round robin order is used to schedule transactions

A high-level overview of the central interconnect crossbar is depicted in Fig. 5. The implementation of the crossbar is based on a mux/demux scheme that requires one multiplexer per Master Mm_AXI ($m = 0, 1, 2, 3, \dots, M-1$, where M is the number of master interfaces) and one demultiplexer per Slave Sn_AXI ($n = 0, 1, 2, 3, \dots, N-1$, where N is the number of slave interfaces).

It should be noted that not all muxes/demuxes have the same number of ports as this depends on the crossbar's connection map. The connection map defines for each slave interface Si_AXI ($i = 0, 1, 2, 3, \dots, N-1$, where i is the index of the slave interface) a set of master port interfaces Mj_AXI ($j = 0, 1, 2, 3, \dots, M-1$, where j is the index of the master interface) that it can access. The connection map is produced as part of the configuration process of the crossbar.

Due to the various functional and performance requirements of the connected accelerators and peripherals, the interconnect supports the tight integration of additional IP blocks in each of its ports. These IP blocks are the following:

1. *AXI Cross Clock IP*: this module implements clock domain crossing for AXI4 transactions, using FIFOs. The use of this module enables AXI Slave ports to operate using different clock frequencies than AXI Master ports. The IP is parameterizable for data and address width.
2. *AXI Data Width Converter IP*: this module implements data width conversion for AXI4 transactions. An AXI Slave port can operate using a different data width than the corresponding AXI Master port, changing the burst length as well. The module is parameterizable for data and address width.

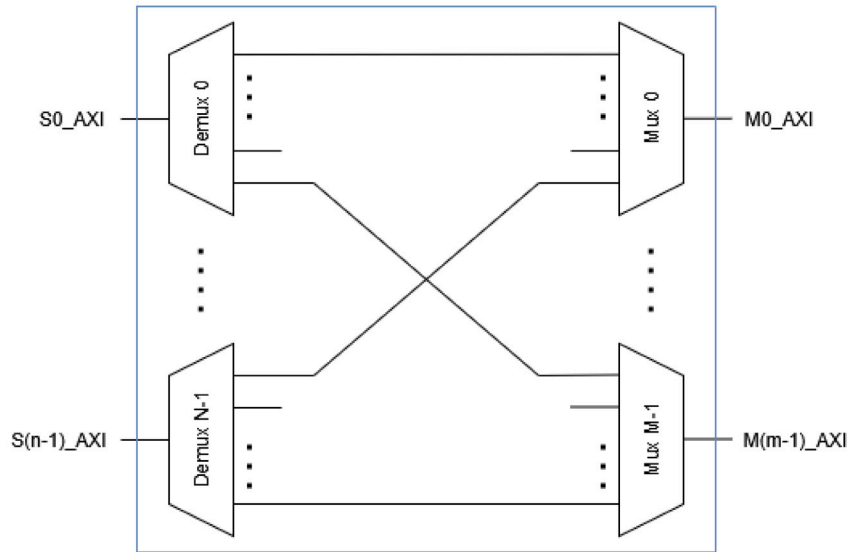


Fig. 5. Overview of the central interconnect crossbar.

3. *AXI Lite to AXI Full IP*: this module implements protocol change from the AXI4-Lite protocol to the AXI4 Full protocol. The module is parameterizable for data and address width.

3.3. Neuromorphic accelerator

The neuromorphic accelerator included in the REBECCA ASIC is a multi-core design based on IMEC's SENECA neuromorphic architecture [14] that has been adapted for the requirements of the REBECCA project. Each neuromorphic core constraints two controllers, a flexible controller (RISC-V/IBEX) and an optimized controller (Loop Buffer), neural processing elements (NPEs) for accelerated network processing, SRAM-based data and instruction memories, and network-on-chip (NoC). The cores are connected using a NoC with sparse and event-driven communication. Fig. 6 provides a high-level overview of the neuromorphic accelerator.

A Neuron core (depicted in Fig. 7) consists of a RISC-V core (based on the open-source IBEX [15], a Neuron Co-Processor, a NoC router, a shared memory interface, and local memory. The RISC-V processor is controlling, programming, and initiating/pausing the Neuron core. The shared memory interface, which is a DMA-like interface that moves data between shared memory and local memory, is also controlled by the RISC-V processor. The Neuron Co-Processor is a programmable SIMD accelerator to compute the states of the neurons and create events when a neuron spikes.

The accelerator configuration used in the actual REBECCA chiplet is comprised of eight such Neuron Cores. Each Neuron core is configured with eight Neuron Processing Engines (NPEs). The data memory included in each core has a size of 2Mb.

3.4. Near-memory computing co-processor

The Near-Memory Computing (NMC) architecture for the REBECCA platform is based on the design presented in [16], where four memory banks in a cache are repurposed, including custom processing units, for the acceleration of neural networks. The architecture of an NMC unit is shown in Fig. 8. Each unit accommodates a matrix of Processing Elements (PEs), and two intermediate buffers to increase data re-usability and reduce the total energy spent on memory accesses. The memory structure is distributed in multiple macros, increasing bandwidth for data intensive applications. A bus module is included for communication with the AXI and a register-mapped control is used for the configuration of the NMC unit. Parallelization of larger workloads

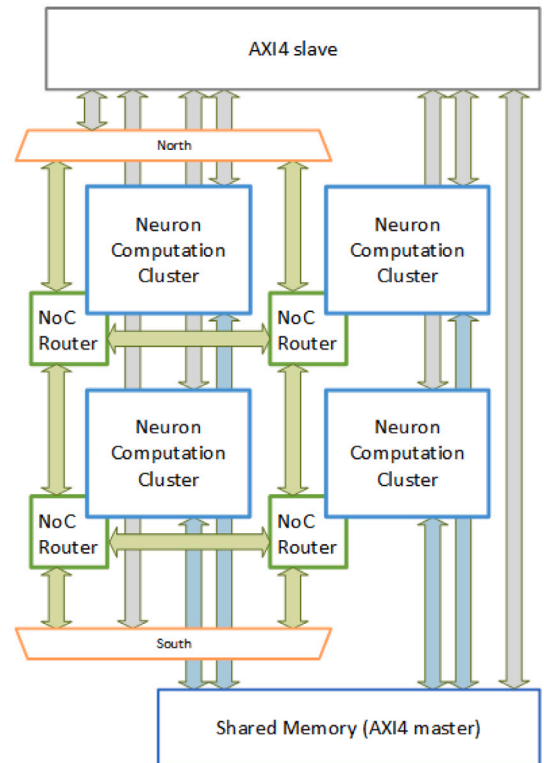


Fig. 6. Simplified overview of the neuromorphic accelerator.

is achieved by distributing computations between all units. For an increased performance gain and memory storage efficiency, the NMC unit supports an 8-bit, per-channel quantization scheme, which is in line with major ML frameworks, e.g., Tensorflow and Open Neural Network Exchange (ONNX).

The NMC implementation for the REBECCA platform also includes a Direct Memory Access (DMA) unit to facilitate data transfer directly from memory to accelerator. The DMA includes address translation to transform the data to and from the system memory into an accelerator-specific memory layout. This reduces the computational overhead in

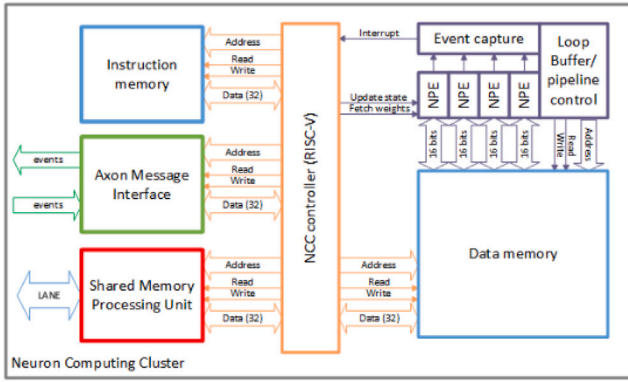


Fig. 7. High-level architecture of the neuron core.

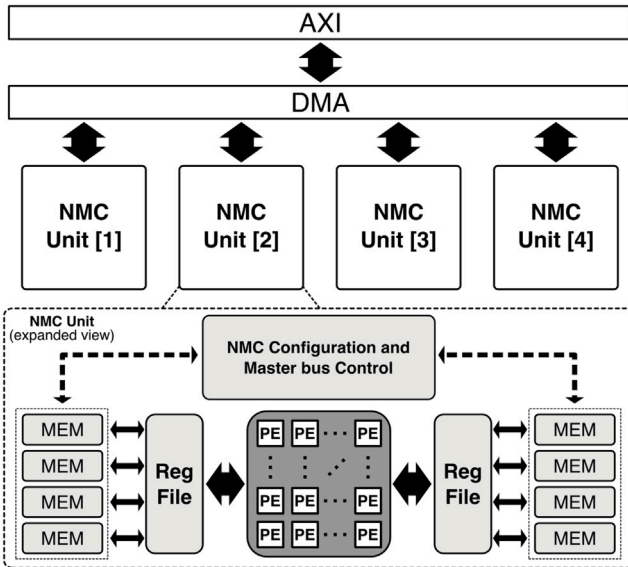


Fig. 8. Near-memory computing architecture overview.

the processor and allows for a accelerator agnostic view of the data layout from the CPU.

The proposed NMC architecture provides optimized hardware acceleration for Edge AI applications integrated in a platform sustaining different workloads with diverse requirements.

3.5. Hardware memory IDS security module

In addition to integrating IP blocks focused on accelerating AI-related compute tasks, the REBECCA ASIC includes a security module that monitors specific memory regions for potentially malicious data, instructions and OS operating patterns. The security module operates similarly to an Intrusion Detection System (IDS) commonly used in network security by comparing data streams accessed from prespecified memory regions against signatures of known malicious vectors.

The security module is based on a pattern-matching algorithm (Aho-Corasick) and focuses on identifying and flagging unwarranted activities that may have infiltrated the REBECCA system. Depending on the operating environment and security policies, the module enables security administrators to specify a body of reference data and signatures of known malicious vectors. From these input data, a series of patterns are created and stored in the internal memory. Once the system resumes operation, when data residing within the memory regions under investigation are accessed, they are accessed by the security module

REBECCA IDS Module

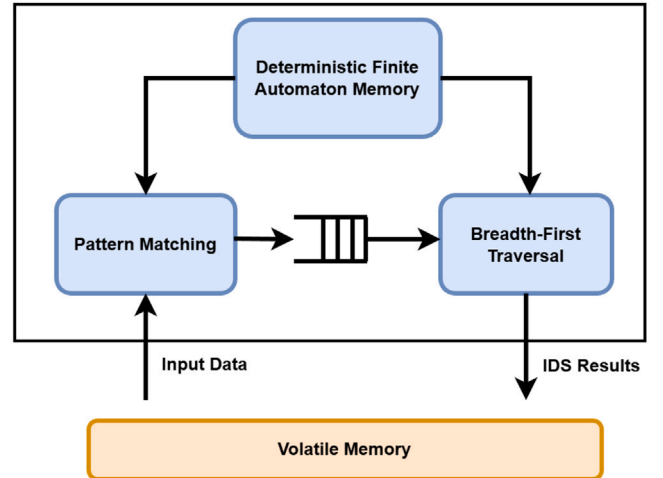


Fig. 9. The REBECCA IDS functional blocks.

and compared against the reference patterns. In the event of a match, the unit notifies the system and appropriate mitigation actions can then be activated.

Fig. 9 demonstrates the high-level functional blocks that comprise the REBECCA IDS mechanisms.

3.6. Peripheral and interface blocks

As the first iteration of the REBECCA platform targets edge devices, various energy, performance and area constraints (including pin counts and PCB footprint) have been considered. To this end, a high-performance alternative to typical DDR memory has been sought. The adoption of HyperRAM and HyperFLASH modules to enable external (volatile and nonvolatile) memory resources has been decided, and as such the internal memory subsystem is complemented by two HyperBus memory controllers connected to pseudoSRAM. REBECCA adapted the HyperRAM memory controller provided by the PULP project [17] by adding support for its central AXI4 interconnect, enabling all connected processors and accelerators to access the memory through the memory controller. Additionally, an SD card controller has also been integrated. The popular non-volatile memory medium is expected to provide storage resources for both OS and data and it can be used as a boot device.

Last but not least, the chiplet integrates two custom communication interfaces. The first one (termed Chip2Chip in Fig. 2) is a communication protocol and physical connection that enables data transfer between the two chiplets of the REBECCA ASIC. The interfacing protocol implemented as a Bunch of Wires (BoW) is a parallel interface that ensures reliable, efficient and high speed communication, supporting data rates ranging from 2 Gbps to 32 Gbps. It supports source synchronous double data rate interfacing.

In the REBECCA ASIC, the implementation of the transceiver system (as shown in Fig. 10) adheres with the BoW standard. A single BoW slice comprises of 18 or 20 bumps that facilitate connections between the ICs. Therefore, a transmitter (Tx) and a receiver (Rx) slice consist of 16 transmitters and 16 receivers respectively. Each transmitter consists of a multiplexer and a driver, while each receiver includes a differential to single-ended converter for the clock, a clock and data synchronizer and a demultiplexer. Additionally, the transceiver system features clock dividers and a PLL.

The second custom interface (Chip2FPGA in Fig. 2) enables data transfers between the REBECCA chiplets and an external FPGA. The

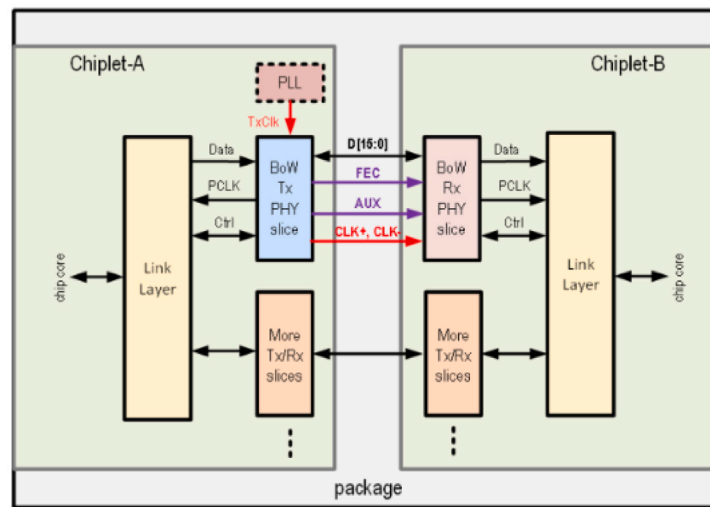


Fig. 10. The Chip2Chip interface blocks connecting the REBECCA chiplets.

interfacing protocol ensures reliable and efficient communication supporting data transfer rates of 100 MHz for each lane. The protocol is based on a modification of Bunch of Wires (BoW) to support communication via the GPIO.

Data transfers from the AXI interface involve several stages. Initially, the AXI data is encapsulated into Transaction Layer Packets (TLPs). Each TLP consists of a header and a payload, with the payload containing the AXI data. The TLP header includes essential control information, such as the TLP type, stream, and auxiliary data.

Once the TLP is formed, it undergoes error protection through Single Error Correction Double Error Detection (SECDED) ECC. Overall, the TLP is 12 bits long. The TLP header and segments of the payload are encoded into a 32-bit small codeword, while larger segments of the payload are encoded into 128-bit large codewords. This encoding ensures that any single-bit errors can be corrected, and double-bit errors can be detected during transmission.

The protected TLP is then quantized into 32-bit granules. These granules are packed into Link Layer Packets (LLPs). An LLP consists of a 32-bit header and 15 granules (480 bits), making the total size 512 bits. The LLP header contains information about the start of TLPs within the LLP and error-checking codes for the header. During transmission, the fragments are serialized and transmitted in the form of granules. The LLPs are sent through the link-physical interface, where the data is serialized to a 16x ratio and then transferred via GPIOs to the receiver. On the receiving end, the data received from the GPIOs is first deserialized. The link layer RX interface then uses the deskew FIFO to align the received fragments correctly. It decodes the LLP header to identify the locations of TLP headers within the granules. The granules are then unpacked from the LLP, and the ECC is used to correct any single-bit errors in the TLPs. Finally, the depacketized TLP payload, which contains the original AXI data, is delivered to the system interface (Chip Core or FPGA processing system). Here, the AXI data is reconstructed, completing the data transfer process from the AXI interface. This multi-stage process ensures data integrity and efficient transfer between the chip and FPGA. In its current stage of development, the Chip2FPGA interface is able to handle transmissions of up to 1.6Gbps.

4. Implementation and testing

The overall REBECCA ASIC system is designed to be implemented in the Global Foundries 22FDX ASIC process. Before reaching this stage, a number of initial prototypes have been designed and implemented as vehicles for testing, verification and validation purposes for all

hardware IPs, while they serve as software bring up platforms. To test and validate different features, four FPGA platforms have been used. The Genesys2 [18] development board has been the main testing platform for the CVA6 core bring up as well as testing with the external HyperRAM devices. The much larger AMD/Xilinx VCU128 [19] and ALINX XKU15 development boards have been used to test the actual multicore system along with different accelerator cores, external memory modules as well as chip to chip and chip to FPGA communications through the extended FMC modules. Lastly, the Alveo U55C [20] server FPGA board has been used as the main software bring up platform, since it enables a convenient way to build and dispatch software OSes and applications on the actual FPGA system shared by multiple developers remotely. For the latter, the initial design was based on the MEEP project [21], which provided essential tools for programming the FPGA and data transfers using the PCIe interface.

On these hardware prototype platforms, the REBECCA consortium has managed to boot Linux OS directly from an SD Card and using a custom designed HyperRAM module consisting of eight HyperRAM chips in two channels and 512MB overall memory capacity. The basic OS is a stripped down version of Linux, with a number of modification in the mainline Linux kernel to support certain features of the hardware platform. Apart from these changes, the OS supports all basic services, including a complete network stack. The latter manages the Ethernet network interfaces provided by the development platforms which have been added to the prototype systems to enable remote access.

Besides the basic Linux OS, the consortium has also managed to boot PikeOS and on top of it PikeOS Native and ElinOS operating systems. Additionally, the Xtratum Hypervisor has been ported and libraries like the EDDL are already running on top of the prototypes. The KlepsydraAI framework is also a work in progress and initial results demonstrate that it is expected to achieve on the final platform, a 10x higher performance over TensorFlow Lite, while being able to successfully execute complex models that standard frameworks could not support.

5. Conclusions

The increasing adoption of AI components in all sorts of applications is currently hindered by the requirements to use cloud-based computations. Performing the necessary computations at the edge is the primary solution to alleviate latency and network connectivity issues as well as security and privacy concerns. REBECCA proposes a complete top-to-bottom solution, covering both hardware and software components, that will enable the effective execution of AI tasks at the edge. The REBECCA platform is built around a chiplet-based SoC that tightly

integrates RISC-V CPU cores along with a series of accelerators and hardware security components that aim to efficiently handle (both in terms of raw performance and in terms of energy efficiency) all demanding AI-related computations. On top of this platform, a complete optimized software stack is also being developed, along with a novel design exploration tool for system and application co-optimization. Together these three main pillars are expected to produce a reference platform that will provide a paradigm shift for edge-AI systems.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Alex Allford reports financial support was provided by Lund University Faculty of Engineering. Masoud Nouripayam reports financial support was provided by Lund University Faculty of Engineering. Arturo Prieto reports financial support was provided by Lund University Faculty of Engineering. Ioannis Papaefstathiou reports financial support was provided by Exascale Performance Systems. Sotiris Ioannidis reports a relationship with Technical University of Crete that includes: employment. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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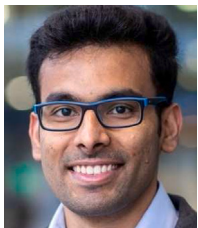
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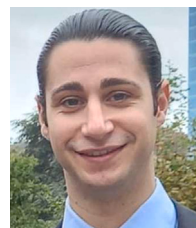
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